

Arulselvan Madhavan

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EDUCATION SUMMARY

Northeastern University, Boston, MA.
Masters in Computer Science(GPA: 3.8)

Sep 2014 - Dec. 2016

TECHNICAL KNOWLEDGE

Languages	: Scala, Go, Python, OCaml, Javascript, C++, C, Haskell, Fortran
Parallel Frameworks	: OpenMP, MPI, GPU - CUDA
Visualization Frameworks	: react, vega, panel, holoviews, bonsai
Performance Tools	: pprof, likwid, perf, flamegraph and HPC@Cori

PROFESSIONAL EXPERIENCE

Machine Systems Engineer at LightMatter, Mountain View

July 2022 - Present

Tools Used: Pytorch, Python, Ocaml

- ML Systems modeling: Built an analytical and event driven simulator for modeling data center level ML training and inference performance incorporating optical interconnects
- Primary contributor to a pytorch extension library that simulated analog matrix multiplications on the photonic hardware with adaptive quantization techniques. [Published in Nature magazine](#)
- Built a [library](#) for reduced bit width quantization(FP8, VS-quant) techniques with calibration support
- Built an [open source library](#) for writing large-scale python source-code transformations in OCaml
- Implemented [mini-dalle](#) and [Stable Diffusion](#) in OCaml

Performance Engineer at Chartboost, San Francisco

Sep 2020 - July 2022

Tools Used: Golang, Scala, Spark, Tensorflow

- Optimized ML Inference time(linear, DNN) with compression, optimized hash maps to be under 20ms.
- Built a low latency real time bidder in Golang that handles 500K requests/sec in under 30 ms
- Built a distributed key-value store that housed terabytes of data with read latency under 10ms
- Built ML training pipelines for data scientists to train and deploy models(linear, DNN) to production
- Built performance(CPU & Memory) observability tools to catch performance regressions in builds

Senior Software Engineer at Chartboost, San Francisco

March 2018 - Sep 2020

Tools Used: Scala, cats, cats-effect, shapeless, refined

- Built a high throughput AdExchange that conducts real time auctions for ad requests(100K /sec)
- Built a low latency, real time ad scoring service by working with data scientists. Latency: 50ms/req
- Led the efforts to promote typed functional programming, type-level programming and category theory among engineers within Chartboost. [Engineering Blog](#)

Software Engineer 2 at Intuit, Mountain View

January 2017 - March 2018

- Designed and implemented a way to encode US tax regulations for all 50 states. Currently, used by more than 200K QuickBooks subscribers while onboarding employees into payroll

Software Developer Intern at Intuit, Mountain View

June 2016 - Aug 2016

- Front end developer for Quickbooks' payroll product responsible for conducting UX research and building responsive user interfaces

Software Developer Co-op at Intuit, Cambridge

July 2015 – Dec 2015

- Primary backend developer responsible for developing web services that connect to various non-relational data sources and enables clients to run "SQL" queries against them

Systems Engineer at Tata Consultancy Services, India

Dec 2011 – Aug 2014

- Primary developer of internal tools team responsible for researching and developing software tools that automate the manual task of engineers

OPEN SOURCE CONTRIBUTION

- [Opine](#) - Python AST unparser written in OCaml. Enables python source code transformations in OCaml
- [Stable Diffusion 1.4](#) - OCaml implementation of stable-diffusion
- [mini-dalle](#) - OCaml implementation of mini-dalle inference path
- [Tsunami Simulation](#) - GPU Programming(OpenMP + OpenMPI) in a HPC @ Cori
- [Category Theory For Programmers - OCaml edition](#) - Primary contributor
- [Scala-steward](#) - Contributed the ability to run this bot on private repos
- [YouTube Channel for scala talks](#)